■ Docker Commands for Beginners – Detailed Guide

This guide introduces essential Docker commands every beginner should know, grouped by category and explained with practical examples.

1. Docker Basics

```
docker --version
```
Check the installed Docker version.

```
docker info
```
Show detailed system-wide information about Docker.

```
docker help
```
List Docker commands or get help for a specific one.

2. Working with Images

```
docker pull
```
Download an image from Docker Hub. Example: docker pull ubuntu

```
docker images
```
List all downloaded images.

```
docker rmi
```
Remove an image. Example: docker rmi ubuntu

```
docker build -t .
```
Build an image from a Dockerfile in the current directory. Example: docker build -t myapp .

3. Running Containers

```
docker run
```
Run a container from an image. Example: docker run ubuntu

```
docker run -it bash
```
Run interactively with a terminal. Example: docker run -it ubuntu bash

```
docker run -d
```
Run a container in the background. Example: docker run -d nginx

```
docker run -p :
```
Map host and container ports. Example: docker run -p 8080:80 nginx

```
docker run -v :
```
Mount a local volume. Example: docker run -v $(pwd):/app node

4. Managing Containers

```
docker ps
```
List running containers.

```
docker ps -a
```
List all containers (including stopped).

```
docker stop
```
Stop a running container.

```
docker start
```
Start a stopped container.

```
docker restart
```
Restart a container.

```
docker rm
```
Remove a container.

5. Inspecting Containers

```
docker logs
```
View container logs.

```
docker exec -it bash
```
Open a shell inside a running container.

```
docker inspect
```
Show detailed container info in JSON.

```
docker top
```
Show running processes in a container.

6. Cleaning Up

```
docker system prune
```
Remove all unused data, images, and stopped containers.

```
docker image prune
```
Remove unused images.

```
docker container prune
```
Remove stopped containers.

```
docker volume prune
```
Remove unused volumes.

7. Docker Compose (Multi-container Apps)

```
docker compose up
```
Start all services defined in docker-compose.yml.

```
docker compose up -d
```
Start in detached mode.

```
docker compose down
```
Stop and remove containers, networks, and volumes.

```
docker compose ps
```
List running compose services.

8. Pro Tips

Use --name

Name your containers for easier management. Example: docker run --name my-nginx -p 8080:80 nginx

Follow logs live

Use docker logs -f to stream logs in real-time.

Use sh for minimal containers

If bash is unavailable, try docker exec -it sh